

Bayesian Optimization Approaches for Constrained Chemical Mixture Optimization

1 Background & Problem Statement

We consider a hydrogen production reaction involving up to 25 chemicals, each of which can have discrete levels ranging from 0 to 5 with a step size of 0.25 for the liquids and 0.20 for the solids. We have the following constraints:

- Cons1:** The number of nonzero components in the mixture must be at most 10.
- Cons2:** The mixture must include at least one light absorber.
- Cons3:** No mixing of acid and base in the same mixture (i.e., if one acid is present, no base can be present, and vice versa).
- Cons4:** The total volume of liquids in the mixture must not exceed 5 mL.
- Cons5:** The total mass of solids in the mixture must not exceed 5 mg.

We denote a single experiment (mixture) by a vector $\mathbf{x} = (x_1, x_2, \dots, x_{25})$, where each x_i can be zero (absent) or in $\{0.25, 0.5, \dots, 5\}$ for liquids (resp. $\{0.20, 0.40, \dots, 5\}$ for solids) if present. Our objective is to maximize the hydrogen yield, denoted $f(\mathbf{x})$, subject to the constraints above and limited experimental budget. We propose two Bayesian Optimization (BO) strategies:

- **Part I: Flattened Approach with All Constraints.**
- **Part II: Two-Level Approach (Presence/Absence + Level) with Split Constraints.**

Both strategies are done in a *batch* manner, where we propose multiple new experiments each iteration.

2 Part I: Flattened Approach with All Constraints

2.1 Flattened Representation

We treat $\mathbf{x} \in \mathbb{R}^{25}$ as a single, 25-dimensional discrete vector, where each component x_i (liquid volume or solid mass) can take values in $\{0, 0.25, 0.5, \dots, 5\}$. In principle:

$$x_i = 0 \quad \text{means the } i\text{-th chemical is absent,}$$

and

$$0.25 \leq x_i \leq 5 \quad \text{means the } i\text{-th chemical is present at that volume/mass level.}$$

We incorporate the following constraints simultaneously:

- **Cons1:** $\sum_{i=1}^{25} \mathbf{1}[x_i > 0] \leq 10$.
- **Cons2:** At least one x_i must correspond to a light absorber chemical (i.e. ≥ 1 absorber is present).
- **Cons3:** Acid/base incompatibility: if i is an acid and j is a base, they cannot both be nonzero.
- **Cons4:** If i indexes a liquid chemical, $\sum_{i \in \text{Liquids}} x_i \leq 5 \text{ mL}$.
- **Cons5:** If i indexes a solid chemical, $\sum_{i \in \text{Solids}} x_i \leq 5 \text{ mg}$.

2.2 Batch Bayesian Optimization with Constraint Handling

Let $f(\mathbf{x})$ be the unknown hydrogen yield. We model f with a surrogate (Gaussian Process) trained on an initial dataset $\mathcal{D}_{\text{init}} = \{(\mathbf{x}^{(j)}, y^{(j)})\}_{j=1}^{n_{\text{init}}}$ which gives both a predicted mean $\mu(\mathbf{x})$ and a standard deviation $\sigma(\mathbf{x})$ for yield at each point $\mathbf{x}$.

We carry out **batch** optimization at each iteration:

1. **Update Surrogate:** Given the current dataset $\mathcal{D}$, update the model $\hat{f}$.
2. **Acquisition Function:** i.e, Expected Improvement
3. **Constrained Maximization (Batch):**
 - Enforce all constraints (Cons1–Cons5).
 - Find a *batch* $\{\mathbf{x}_1, \dots, \mathbf{x}_B\}$ maximizing $\alpha(\mathbf{x})$ subject to these constraints.
4. **Experiment:** Evaluate $f(\mathbf{x}_1), \dots, f(\mathbf{x}_B)$ in the lab.
5. **Update Dataset:** Augment $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{x}_b, f(\mathbf{x}_b)) : b = 1, \dots, B\}$.

We iterate until the budget is exhausted. Because we keep the full 25-dimensional vector $\mathbf{x}$, the surrogate may capture synergy between presence/absence and level in a single model.

3 Part II: Two-Level Approach (Presence/Absence + Level)

Here, we split the problem into two sequential steps: (i) selecting which chemicals to include (presence/absence) and (ii) optimizing the levels of those selected chemicals. We also *split the constraints* so that the first three constraints (Cons1, Cons2, Cons3) govern presence/absence, while the last two constraints (Cons4, Cons5) govern the total volumes/masses in the level optimization step. **Consider Auto-Encoders, but we can start with this simpler approach and see what we get.**

3.1 Stage 1: Subset Selection via Random Forest (Cons1, Cons2, Cons3)

We first ignore volume/mass constraints (Cons4, Cons5) and only apply:

- **Cons1:** Up to 10 chemicals,
- **Cons2:** At least one light absorber,
- **Cons3:** No simultaneous acid/base in the subset.

3.1.1 Presence Encoding

Convert $\mathbf{x} = (x_1, \dots, x_{25})$ into a *binary* vector $\mathbf{z} \in \{0, 1\}^{25}$ via

$$z_i = \begin{cases} 1, & \text{if } x_i > 0, \\ 0, & \text{otherwise.} \end{cases}$$

From an initial dataset $\mathcal{D} = \{(\mathbf{x}^{(j)}, y^{(j)})\}$, form $\mathcal{D}_z = \{(\mathbf{z}^{(j)}, y^{(j)})\}$ and train a **random forest** model $g(\mathbf{z})$ that predicts yield from $\mathbf{z}$ alone. We also estimate an uncertainty $\sigma_g(\mathbf{z})$ from the forest (e.g., the standard deviation among the ensemble of trees).

3.1.2 Batch Subset Acquisition

At each iteration:

1. **Update g :** Refit the RF model $g(\mathbf{z})$ on $\mathcal{D}_z$.
2. **Acquisition Function:** $\alpha_g(\mathbf{z})$ could be *Upper Confident Bound (UCB)*

$$\alpha_g(\mathbf{z}) = \mu_g(\mathbf{z}) + \kappa \sigma_g(\mathbf{z}),$$

where $\mu_g(\mathbf{z})$ is the mean prediction of g , and $\sigma_g(\mathbf{z})$ its uncertainty measure.

3. **Constraint Enforcement (Stage 1):**

- $\|\mathbf{z}\|_1 = \sum_{i=1}^{25} z_i \leq 10$ (Cons1).
- At least one z_i must indicate a light absorber (Cons2).
- No z_i and z_j can both be 1 if i is acid and j is base (Cons3).

4. **Select Batch of Subsets:** Maximize $\alpha_g(\mathbf{z})$ subject to the above presence constraints, yielding $\{\mathbf{z}_1, \dots, \mathbf{z}_B\}$.

These $\mathbf{z}_b$ indicate which chemicals to include but not at what level.

3.2 Stage 2: Level Optimization via GP (Cons4, Cons5)

Next, we incorporate the volume/mass constraints (Cons4, Cons5). For each chosen subset $\mathbf{z}_b$ in the batch from Stage 1:

1. **Update GP:** Fit a Gaussian Process $f(\mathbf{x})$ to the entire dataset $\mathcal{D} = \{(\mathbf{x}, y)\}$, i.e. the original level-based data.
2. **Restricted Level Space:** For the chemicals indicated by $\mathbf{z}_b$, let $x_i \in \{0.25, \dots, 5\}$ if $z_{b,i} = 1$, and $x_i = 0$ otherwise.
3. **Add Cons4, Cons5:**
 - $\sum_{i \in \text{Liquids}} x_i \leq 5 \text{ mL}$.
 - $\sum_{i \in \text{Solids}} x_i \leq 5 \text{ mg}$.
4. **Level Acquisition (Batch):** Define an acquisition function for the GP, e.g. EI, and maximize it over the feasible region (Cons4–5).
5. **Propose New Mixtures:** For each subset $\mathbf{z}_b$, pick one or more $\mathbf{x}_b$ that maximize $\alpha_f(\mathbf{x})$, subject to the above constraints. Collect them into a single batch $\{\mathbf{x}_1, \dots, \mathbf{x}_B\}$ for the lab.

3.3 Execution and Iteration

Both models (g for presence and f for levels) are updated after new data arrives. Future iterations can discover new subsets if earlier guesses are outperformed. Over multiple iterations, this *two-level* approach can converge on a strongly performing subset and appropriate levels.

4 Comparison and Notes

- **Constraint Allocation:**
 - **Part I (Flattened):** Enforces all constraints (Cons1–5) in a single mixed discrete optimization step.
 - **Part II (Two-Level):** Splits constraints so that presence constraints (Cons1–3) guide subset selection, then level constraints (Cons4–5) are applied once the subset is chosen.
- **Batch Optimization:** Both approaches propose a batch of $B > 1$ new mixtures each iteration, which is often more time-efficient if one can run multiple experiments in parallel.
- **Uncertainty Estimates:**
 - In **Part I**, the GP model for the flattened vector $\mathbf{x}$ yields $\sigma(\mathbf{x})$
 - In **Part II**, the random forest on $\mathbf{z}$ provides $\sigma_g(\mathbf{z})$, and the GP on $\mathbf{x}$ provides $\sigma_f(\mathbf{x})$.
- **Trade-offs:** The flattened approach can capture all interactions in one model but might be harder to optimize in a single step. The two-level approach is more interpretable (first choose the subset, then fine-tune levels) but can miss synergy if presence/absence and level interactions are strong.